

Smart Emergency Route System

A MINI PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT FOR THE AWARD OF
THE DEGREE OF
MASTER
IN
COMPUTER APPLICATIONS

Submitted by

AKASH KUMAR_25MCA20049

Supervisor

Ms. Nisha Sharma_E2915
ASSISTANT PROFESSOR, UIC
CHANDIGARH UNIVERSITY



UNIVERSITY INSTITUTE OF COMPUTING
CHANDIGARH UNIVERSITY
MOHALI, PUNJAB-140301

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ABSTRACT

This project focuses on the design and development of a Smart Emergency Route System that utilizes Artificial Intelligence-based search algorithms to determine the most efficient path for emergency vehicles such as ambulances. In real-life scenarios, reaching a destination quickly during emergencies is critical, and traffic congestion or road blockages can significantly delay response time. This system aims to simulate such real-world conditions and provide an intelligent solution for optimal route selection.

The environment in this project is modeled as a grid-based map where each cell represents a road segment with an associated cost. These costs are used to simulate varying traffic conditions, where lower values represent smoother traffic flow and higher values indicate congestion. Additionally, certain cells can be blocked to represent road closures or obstacles, further enhancing the realism of the system. Users can interact with the system through a graphical user interface (GUI) developed using Python, where they can select the start and destination points and visualize the computed routes.

The system implements two primary search algorithms: Breadth-First Search (BFS) and the A* Search Algorithm. BFS is an uninformed search technique that finds the shortest path based on the number of steps, without considering traffic conditions. In contrast, the A* algorithm is an informed search technique that uses a heuristic function along with actual path cost to determine the most optimal route. This allows A* to make smarter decisions by prioritizing paths with lower estimated total cost.

The performance of both algorithms is analyzed based on metrics such as path length, total cost, and number of nodes explored. The results demonstrate that while BFS is effective in simple scenarios, A* provides more efficient and practical solutions in environments with varying traffic conditions and obstacles. The system also includes features such as dynamic traffic simulation, road blocking, and real-time visualization of the route with an animated ambulance.

Overall, this project highlights the significance of intelligent search algorithms in real-time applications and demonstrates how AI techniques can be applied to improve emergency response systems. It serves as a strong foundation for future enhancements such as integration with real-world maps, live traffic data, and advanced optimization techniques.

CHAPTER 1

INTRODUCTION

1.1 Project Overview

The **Smart Emergency Route System** is an Artificial Intelligence-based application designed to identify the most efficient path for emergency vehicles such as ambulances in a simulated environment. The primary objective of this project is to reduce response time during critical situations by providing an optimized route that considers both distance and traffic conditions.

In real-world scenarios, emergency services often face challenges such as traffic congestion, road blockages, and inefficient navigation systems, which can lead to delays. This project addresses these challenges by modeling a city environment as a grid-based structure, where each cell represents a segment of the road network. Each cell is assigned a cost value that reflects the level of traffic, with higher values indicating heavier congestion and lower values representing smoother traffic flow. Additionally, certain cells can be blocked to simulate road closures or obstacles.

The system incorporates two fundamental search algorithms: **Breadth-First Search (BFS)** and the **A* Search Algorithm**. BFS is used to determine the shortest path based on the number of steps, making it suitable for simple scenarios where all paths have equal cost. On the other hand, A* is a more advanced and efficient algorithm that calculates the optimal path by considering both the actual cost of travel and an estimated cost to the destination using a heuristic function. This enables the system to make intelligent routing decisions in dynamic environments.

An interactive graphical user interface (GUI) is developed using Python to enhance user experience and provide real-time visualization of the system. Users can select the starting point and destination, simulate traffic conditions, block roads, and observe how the algorithms compute and display the route. The system also provides performance metrics such as path length, total cost, and number of nodes explored, allowing for a comparative analysis of algorithm efficiency.

Overall, this project demonstrates how Artificial Intelligence techniques can be effectively applied to solve real-world problems in emergency response systems. It provides a scalable and extensible framework that can be further enhanced with real-time data integration and advanced optimization methods.

1.2 Objective

The main objective of the Smart Emergency Route System is to develop an intelligent solution that can determine the most efficient path for emergency vehicles under varying traffic conditions. The project aims to simulate real-world scenarios and demonstrate the effectiveness of Artificial Intelligence algorithms in optimizing route selection.

The specific objectives of this project are as follows:

- To design and implement a grid-based environment that represents a real-world road network with varying traffic conditions.
- To apply Breadth-First Search (BFS) algorithm for finding the shortest path based on the number of steps without considering traffic cost.
- To implement the A* Search Algorithm for determining the optimal path by considering both actual cost and heuristic-based estimated cost.
- To compare the performance of BFS and A* algorithms using parameters such as path length, total cost, and number of nodes explored.
- To develop an interactive Graphical User Interface (GUI) using Python for better visualization and user interaction.
- To simulate real-world conditions such as traffic congestion and road blockages to enhance the practical applicability of the system.
- To visualize the movement of an emergency vehicle (ambulance) along the computed path for

- better understanding of the routing process.
- To demonstrate how AI-based search algorithms can improve decision-making in time-critical situations like emergency response.

1.3 Motivation

The motivation behind developing the **Smart Emergency Route System** arises from the critical need to improve response times in emergency situations such as medical crises, accidents, and natural disasters. In real-world scenarios, delays in reaching the destination can have serious consequences, including loss of life. One of the major factors contributing to such delays is inefficient route planning due to traffic congestion, road blockages, and lack of intelligent navigation systems.

With the rapid growth of urban populations and the increasing number of vehicles on roads, traffic management has become a significant challenge. Traditional navigation systems often provide routes based only on distance, without effectively considering dynamic factors such as traffic density or unexpected obstacles. This creates a need for smarter systems that can adapt to changing conditions and provide optimal routing solutions in real time.

This project is motivated by the potential of Artificial Intelligence to address such real-world problems. By applying search algorithms like **Breadth-First Search (BFS)** and the **A* Search Algorithm**, it is possible to simulate intelligent decision-making processes that can identify efficient paths under different conditions. The use of heuristics in A* further enhances its ability to provide faster and more practical solutions compared to conventional methods.

Another key motivation is to bridge the gap between theoretical knowledge and practical application. Concepts of search algorithms are often studied in academic courses, but their real-world implementation helps in better understanding and skill development. This project provides a hands-on approach to learning and demonstrates how these algorithms can be applied in domains such as transportation, logistics, and emergency management.

Furthermore, the development of a graphical user interface (GUI) allows users to visually interact with the system, making it easier to understand how different algorithms behave and perform. This enhances both usability and learning experience.

Overall, the motivation for this project is to develop an intelligent, efficient, and practical system that showcases the importance of AI in solving real-life problems, particularly in improving emergency response systems and saving valuable time.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing System

In the current real-world scenario, navigation and route planning for emergency vehicles primarily rely on conventional systems such as GPS-based navigation tools and standard mapping applications. These systems are widely used for determining routes between a source and a destination; however, they have several limitations when it comes to handling emergency situations effectively.

Most existing systems focus mainly on providing the shortest path based on distance or estimated travel time. While some advanced applications incorporate live traffic updates, they often lack the ability to dynamically adapt to sudden changes such as road blockages, accidents, or unexpected congestion in real time. Additionally, these systems do not always prioritize emergency vehicles, which can lead to delays in critical situations.

Another limitation of existing systems is the absence of intelligent decision-making mechanisms. Traditional navigation tools do not utilize advanced Artificial Intelligence algorithms to optimize route selection. Instead, they rely on predefined algorithms that may not always produce the most efficient results in complex or dynamic environments. This becomes a major drawback in emergency scenarios where every second is crucial.

Furthermore, most current systems do not provide a comparative analysis of different routing techniques. Users are generally presented with one or two route options without understanding how those routes are computed or how efficient they are in terms of cost, time, or resource utilization. This lack of transparency limits the ability to evaluate and improve routing strategies.

2.2 Proposed System

The proposed system is a **Smart Emergency Route System** that utilizes Artificial Intelligence-based search algorithms to determine the most efficient path for emergency vehicles in a dynamic environment. The system is designed to overcome the limitations of existing navigation systems by incorporating intelligent decision-making, real-time simulation of traffic conditions, and interactive visualization.

In this system, the environment is represented as a grid-based map where each cell corresponds to a segment of the road network. Each cell is assigned a numerical cost value that reflects the level of traffic congestion. Lower values indicate smooth traffic flow, while higher values represent heavy congestion. Additionally, certain cells can be marked as blocked to simulate road closures or obstacles, making the environment more realistic and dynamic.

The proposed system implements two key search algorithms: **Breadth-First Search (BFS)** and the **A* Search Algorithm**. BFS is used to determine the shortest path based on the number of steps, making it suitable for uniform environments where all paths have equal cost. In contrast, the A* algorithm uses a heuristic function along with actual path cost to calculate the optimal route, enabling it to handle varying traffic conditions efficiently. This allows the system to provide more practical and cost-effective solutions compared to traditional methods.

A user-friendly Graphical User Interface (GUI) is developed using Python, allowing users to interact with the system easily. Users can select the starting point and destination by clicking on the grid, simulate traffic conditions, block roads, and execute the algorithms. The system visually displays the computed path and also animates the movement of an ambulance along the selected route, enhancing the overall user experience.

2.3 Feasibility Study

The feasibility study evaluates the practicality and viability of the **Smart Emergency Route System** in terms of technical, economic, and operational aspects. It helps determine whether the proposed system can be successfully developed and implemented within the given constraints.

1. Technical Feasibility

The proposed system is technically feasible as it is developed using widely available technologies such as Python and its built-in libraries. The implementation of search algorithms like Breadth-First Search (BFS) and A* is straightforward and does not require complex hardware or advanced infrastructure. The system runs efficiently on standard computing devices with minimal resource requirements. Additionally, the graphical user interface (GUI) is developed using simple tools, making it easy to design, implement, and maintain. Therefore, no specialized technical expertise or expensive software is required.

2. Economic Feasibility

The system is economically feasible as it involves minimal development cost. It uses open-source programming tools and libraries, eliminating the need for purchasing licensed software. Since the project is implemented in a simulated environment, there are no additional costs related to hardware installation or real-world deployment. Maintenance and upgrades can also be performed with minimal expense. This makes the project cost-effective and suitable for academic as well as small-scale practical applications.

3. Operational Feasibility

The system is user-friendly and easy to operate. The graphical interface allows users to interact with the system by simply selecting start and destination points and executing algorithms using buttons. No prior technical knowledge is required to use the system effectively. The system provides clear visual outputs and performance metrics, making it suitable for both demonstration and learning purposes. It can be easily integrated into educational environments to demonstrate the working of AI-based search algorithms.

4. Time Feasibility

The development of the system can be completed within a reasonable time frame, as it involves well-defined algorithms and straightforward implementation. Testing and debugging are also manageable due to the modular structure of the system. This makes the project feasible within academic deadlines.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 System Architecture

The Smart Emergency Route System is built using a layered architecture that integrates BFS (Breadth First Search) and A* algorithm for efficient and optimized emergency route calculation. The system is designed to ensure fast response in critical situations by finding the shortest and most reliable path between source and destination locations. It is structured into multiple layers that work together to process user requests and generate optimal routes in real time.

The first layer is the presentation layer, which provides a simple and interactive interface for users to input source and destination locations during an emergency. The second layer is the processing layer, where the main routing logic is implemented. In this layer, the system first uses BFS to explore all possible connected paths in the road network and then applies the A* algorithm to calculate the most optimal route based on distance and heuristic cost estimation. This combination ensures both completeness and efficiency in path selection.

The second layer is the data layer, which contains a graph-based representation of the entire road network. Here, intersections are treated as nodes and roads as weighted edges, where weights may represent distance, travel time, or traffic conditions. This layer also supports dynamic updates such as traffic congestion or road blockages. Finally, the system processes all inputs and generates the fastest and most efficient emergency route, helping reduce response time in critical scenarios.

3.2 Module Description

The Smart Emergency Route System is divided into several functional modules that work together to efficiently compute and deliver the optimal emergency route. Each module is responsible for a specific task, ensuring smooth flow of data and accurate route generation in real time.

The first module is the User Interface Module, which allows users to input the source and destination locations in an emergency situation. This module is designed to be simple, fast, and easy to use so that users can quickly request a route without confusion or delay. It also displays the final optimized path generated by the system.

The second module is the Graph Construction Module, which converts the road network into a graph structure. In this module, intersections are represented as nodes and roads are represented as edges with assigned weights such as distance, travel time, or traffic density. This structured representation helps in efficient pathfinding and data processing.

The third module is the Pathfinding Module, which is the core of the system. It first uses BFS to explore all possible routes and ensure connectivity, and then applies the A* algorithm to calculate the most optimal path based on heuristic and actual cost values. This module ensures that the system finds the shortest and fastest emergency route.

The fourth module is the Optimization and Decision Module, which evaluates multiple possible routes and selects the best one based on real-time conditions such as traffic congestion or road closures. This module improves the accuracy and reliability of the system.

The final module is the Output and Visualization Module, which displays the selected optimal route to the user in a clear and visual format. It ensures that users can easily understand the path and follow it during emergency situations.

3.3 Technology Stack

The Smart Emergency Route System is developed using a lightweight and efficient technology stack that supports graph-based computation, real-time processing, and optimal pathfinding. The system is designed to ensure fast execution and accurate emergency route generation with minimal complexity.

- **Programming Language:** The system is implemented in Python, as it provides strong support for data structures, graph algorithms, and rapid development. Its simplicity and rich libraries make it suitable for implementing BFS and A* algorithms efficiently.
- **Core Algorithms:** Breadth First Search (BFS) is used for initial exploration of all possible routes in the network, while the A* algorithm is used for optimal path selection using heuristic-based cost estimation. This combination improves both accuracy and performance in emergency routing.
- **Data Structure:** A graph-based data structure is used to represent the road network. In this structure, nodes represent intersections or locations, and weighted edges represent roads with parameters such as distance, travel time, or traffic load.
- **Development Environment:** The system is developed and tested using standard IDEs such as Visual Studio Code or PyCharm, which support debugging, code organization, and efficient execution of Python programs.
- **Visualization Support:** Graph visualization techniques are used to represent the computed optimal route in a clear and understandable format, helping users easily follow the emergency path.
- **System Compatibility:** The application is platform-independent and can run on different operating systems such as Windows, Linux, and macOS without requiring major changes in the codebase.

CHAPTER 4

SYSTEM TESTING AND RESULTS

4.1 Testing Approaches

The Smart Emergency Route System is tested using multiple testing approaches to ensure correctness, reliability, and efficiency of the routing results. Since the system is based on BFS and A* algorithms, testing focuses on validating both functional accuracy and performance under different scenarios.

- **Unit Testing:** Individual components of the system such as graph creation, BFS traversal, and A* path calculation are tested separately to ensure that each module works correctly before integration.
- **Integration Testing:** After combining all modules, integration testing is performed to check the interaction between the user interface, graph processing, and pathfinding modules. This ensures that data flows correctly between components.
- **Functional Testing:** This approach verifies that the system correctly finds the shortest and most optimal emergency route based on the given source and destination inputs. It checks whether BFS and A* produce accurate results.
- **Performance Testing:** The system is tested under different graph sizes and complexities to evaluate its speed and efficiency. This ensures that the route is generated quickly even in large and complex road networks.
- **Boundary Testing:** Edge cases such as disconnected nodes, no available path, or multiple shortest paths are tested to ensure the system handles unusual conditions properly.
- **Validation Testing:** The final output routes are validated to confirm that they are logical, optimal, and suitable for real-time emergency situations.

4.2 Test Cases

TC01: Valid Route Calculation

The system is tested by providing a valid source and destination, and it correctly generates the shortest emergency route using BFS and A* algorithm.

TC02: Same Source and Destination

When the source and destination are the same, the system properly identifies it and displays a message indicating that the user is already at the destination.

TC03: No Path Available

In case of disconnected nodes where no route exists, the system correctly handles the situation and displays “No route found” message.

TC04: Multiple Path Options

When multiple possible routes exist, the system selects the most optimal path based on the lowest cost using the A* heuristic function.

TC05: Large Graph Testing

The system is tested on a large road network with multiple nodes, and it successfully generates the optimal route within acceptable processing time.

TC06: Weighted Graph Scenario

Different road weights such as distance or traffic are applied, and the system correctly selects the least-cost path.

TC07: Invalid Input Handling

When empty or incorrect inputs are provided, the system displays proper validation error messages.

TC08: Dynamic Road Condition Update

In case of road blockage or changes in the network, the system recalculates and provides an alternate optimal route automatically.

4.3 Results

- The Smart Emergency Route System successfully generates the shortest and most optimal path between source and destination using BFS and A* algorithm.
- The system provides fast response time, making it suitable for emergency situations where quick decision-making is required.
- It accurately handles different scenarios such as multiple routes, blocked paths, and weighted graphs by selecting the most efficient route.
- The integration of heuristic-based A* algorithm improves the performance and reduces unnecessary path exploration.
- The system effectively manages large road networks and still maintains acceptable processing speed and accuracy.
- ☐ All test cases are successfully passed, which confirms that the system is reliable, efficient, and suitable for real-world emergency navigation use.



CHAPTER 5

CONCLUSION AND FUTURE WORK

1.1 Conclusion

The Smart Emergency Route System effectively achieves its primary goal of providing an optimized and reliable emergency routing solution using BFS and A* algorithms. The system is capable of analyzing complex road networks and determining the shortest and most efficient path between source and destination in a minimal amount of time, which is crucial in emergency situations.

The hybrid approach of BFS for exploring possible paths and A* for heuristic-based optimization enhances both accuracy and performance. This combination ensures that the system not only finds a valid route but also selects the most optimal one based on real-time cost factors such as distance and traffic weight. As a result, the system improves response time and reduces delays during critical emergencies.

Furthermore, the system demonstrates strong adaptability in handling various real-world conditions such as multiple route options, disconnected nodes, and dynamic changes like roadblocks or traffic variations. The testing results confirm that the system is stable, efficient, and capable of performing well under different scenarios.

In conclusion, this project provides a strong foundation for intelligent emergency navigation systems. With future enhancements such as real-time GPS tracking, live traffic integration, and mobile application support, the system can be further improved to deliver even more accurate and real-time emergency assistance.

1.2 Future Enhancement

The Smart Emergency Route System has strong potential for further improvements to make it more efficient and suitable for real-world applications. With advancements in technology, the system can be upgraded to provide more accurate, real-time, and intelligent routing solutions. These enhancements will help in improving performance, reliability, and user experience during emergency situations. Future updates can also make the system more scalable and adaptive to dynamic environmental conditions. Overall, these improvements will make the system more powerful and practical for real-time emergency navigation.

- The system can be improved by integrating real-time traffic data to adjust route weights dynamically and provide more accurate emergency paths.
- GPS-based live tracking can be added to monitor the real-time location of users and emergency vehicles for better navigation accuracy.
- A mobile application version can be developed to make the system more accessible and user-friendly during emergency situations.
- Machine learning techniques can be implemented to predict traffic conditions and suggest more optimized routes in advance.
- Voice-assisted navigation can be added to help users in situations where manual interaction is not possible.
- The system can be integrated with emergency services like ambulances, police, and fire departments for faster response and coordination.
- Cloud-based processing and storage can be used to improve scalability and handle large road network data efficiently.

REFERENCES

- GeeksforGeeks — BFS Algorithm
<https://www.geeksforgeeks.org/breadth-first-search-bfs-for-a-graph/>
- GeeksforGeeks — A* Search Algorithm
<https://www.geeksforgeeks.org/a-search-algorithm/>
- Tutorialspoint — Breadth First Search (BFS)
<https://www.tutorialspoint.com/breadth-first-search-or-bfs-for-a-graph>
- NPTEL (IIT India) — Artificial Intelligence & Graph Search Lectures
<https://nptel.ac.in/>
- GeeksforGeeks — Graph Theory Basics
<https://www.geeksforgeeks.org/graph-data-structure-and-algorithms/>
- Tutorialspoint — Dijkstra and Shortest Path Concepts
https://www.tutorialspoint.com/data_structures_algorithms/dijkstra_algorithm.htm

GitHubLink:- <https://github.com/akashkumar10596/Smart-Emergency-Route-System>